

**Q1.**

A small, hollow, plastic ball of mass  $m$  kg is at rest on a point  $O$  on a polished horizontal surface. The ball is attached to two identical springs. The other ends of the springs are attached to the points  $P$  and  $Q$  which are 1.8 metres apart on a straight line through  $O$ .

The ball is struck so that it moves away from  $O$ , towards  $P$  with a speed of  $0.75 \text{ m s}^{-1}$ .

As the ball moves, its displacement from  $O$  is  $x$  metres at time  $t$  seconds after the motion starts.

The force that each of the springs applies to the ball is  $12.5mx$  newtons towards  $O$ .

The ball is to be modelled as a particle. The surface is assumed to be smooth and it is assumed that the forces applied to the ball by the springs are the only horizontal forces acting on the ball.

- (a) Find the minimum distance of the ball from  $P$ , in the subsequent motion.

(5)

- (b) In practice the minimum distance predicted by the model is incorrect.

Is the minimum distance predicted by the model likely to be too big or too small?

Explain your answer with reference to the model.

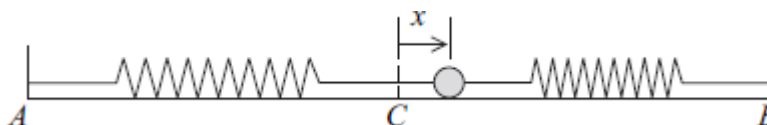
(2)

(Total 7 marks)

**Q2.**

A particle, of mass 9 kg, is attached to two identical springs. The other ends of the springs are attached to fixed points,  $A$  and  $B$ , which are 1.2 metres apart on a smooth horizontal surface. The springs have modulus of elasticity 45 N and natural length 0.4 m.

The particle is released from rest at a distance of 0.5 metres from  $B$  and moves on the line  $AB$ . The midpoint of  $AB$  is  $C$ . At time  $t$  seconds after release, the displacement of the particle from  $C$  is  $x$  metres, where the direction from  $A$  to  $B$  is taken to be positive.



- (a) Show that the resultant force on the particle, at time  $t$ , is  $-225x$  newtons.

(4)

- (b) Hence show that the particle moves with simple harmonic motion.

(2)

- (c) State the period of this motion.

(2)

- (d) Find the speed of the particle when it is 0.05 metres from  $C$ .

(4)

- (e) Write down an expression for  $x$  in terms of  $t$ .

(2)

(Total 14 marks)

### Q3.

A particle  $P$ , of mass  $0.4\text{ kg}$ , is attached to one end of a light elastic string, and the other end of the string is attached to a fixed point  $A$ .

- (a) The particle  $P$  hangs in equilibrium at a point  $E$ , vertically below  $A$ , where the extension of the string is  $0.2\text{ metres}$ . Calculate the stiffness of the string.

(3)

- (b) The particle  $P$  is pulled vertically downwards from the point  $E$  by a distance of  $0.1\text{ metres}$ , and released from rest. The displacement of  $P$  from  $E$  at time  $t$  seconds after being released is  $x\text{ metres}$ .

- (i) Given that the string does not become slack during the subsequent motion, show that

$$\ddot{x} = hx$$

where  $h$  is a constant to be determined.

(4)

- (ii) Hence deduce that the motion of  $P$  is simple harmonic.

(1)

- (iii) Show that the period of this motion is  $0.898\text{ seconds}$ , correct to three significant figures.

(2)

- (iv) Calculate the maximum speed of  $P$  during its motion.

(2)

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(Total 12 marks)

### Q4.

A railway truck, of mass  $m$ , is travelling in a straight line along a horizontal track.

At time  $t = 0$ , the truck strikes one end of a buffer which is fixed at its other end.

The buffer may be modelled as a light spring of natural length  $a$  and modulus of elasticity  $am n^2$ , where  $n$  is a positive constant. At time  $t$ , the compression of the buffer is  $x$ .

- (a) In a simple model of the motion, the only force affecting the truck during this motion is the thrust from the buffer.

- (i) Show that, while the truck is in contact with the buffer, the truck performs simple harmonic motion.

(2)

- (ii) Find, in terms of  $n$ , the period of this motion.

(1)

- (b) In a more realistic model, the motion of the truck is affected by a resistance force of magnitude  $mkv$ , where  $v$  is the speed of the truck and  $k$  is a positive constant.

- (i) Show that, while the buffer is being compressed,  $x$  satisfies the equation

$$\ddot{x} + K\dot{x} + n^2x = 0$$

(3)

- (ii) At time  $t = 0$ , the truck is travelling with speed  $U$ . Given that  $k = \frac{5n}{2}$  find  $x$  in terms of  $n$ ,  $U$  and  $t$ .

(6)

- (iii) By means of a sketch, or otherwise, explain whether the type of damping is light, critical or heavy.

(2)

(Total 14 marks)

### Q5.

A particle moves along a straight line between the points  $A$  and  $B$  with simple harmonic motion. The point  $O$  is the mid-point of  $AB$ . At time  $t$  seconds, the particle is  $x$  metres from  $O$  and moving with speed  $v \text{ m s}^{-1}$ .

The motion of the particle satisfies the equation

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

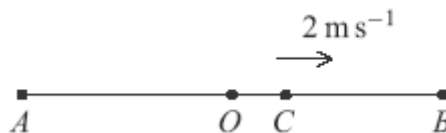
- (a) The particle completes one oscillation in 7.5 seconds.

Show that  $\omega = \frac{4\pi}{15}$ .

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(2)

- (b) When the particle passes through the point  $C$ , as shown in the diagram,  $x = 1$  and  $v = 2$ .



- (i) Show that the amplitude of the motion is 2.59 metres, correct to three significant figures.

(2)

- (ii) When the particle first passes through  $C$ , it is heading away from  $O$  towards  $B$ .

Find the time that it takes to move from  $C$  to  $B$  and back to  $C$ , giving your answer to two significant figures.

(5)

- (iii) Find the maximum speed of the particle during the oscillations.

(2)

(Total 11 marks)

**Q6.**

A particle moves in a straight line with simple harmonic motion such that its displacement at time  $t$  seconds relative to a fixed origin on this line is  $x$  metres. The motion of the particle satisfies the differential equation

$$\frac{d^2 x}{dt^2} + 16x = 0$$

- (a) Verify that

$$x = A \cos 4t + B \sin 4t$$

where  $A$  and  $B$  are constants, is a solution to this differential equation.

(4)

- (b) When  $t = 0$ , the particle is momentarily at rest. Show that  $B = 0$ .

(1)

- (c) Given that  $x = h$  ( $h > 0$ ) when  $t = \frac{\pi}{12}$ , find  $A$  in terms of  $h$ .

(2)

- (d) Find the maximum speed of the particle in terms of  $h$ .

(1)

- (e) The mass of the particle is  $m$  kg. Find the magnitude of the maximum force acting on the particle during the motion. Give your answer in terms of  $h$  and  $m$ .

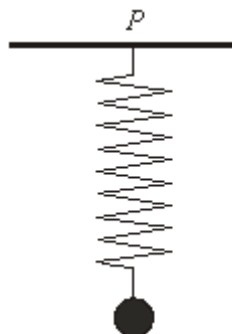
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(Total 10 marks)

**Q7.**

A sphere of mass 2 kg is fixed to one end of a spring. The other end of the spring is attached to the fixed point  $P$ , as shown in the diagram. The natural length of the spring is 0.5 metres.



- (a) When the system is in equilibrium, the length of the spring is 0.7 metres.

Find the modulus of elasticity of the spring.

(3)

- (b) The sphere is pulled down below its equilibrium position and released from rest. At time  $t$  seconds, the displacement of the sphere from its equilibrium position is  $x$  metres.

- (i) Show that

$$\frac{d^2 x}{dt^2} = -49x$$

(5)

- (ii) Find the value of  $t$  when the sphere returns to its equilibrium position for the first time.

(4)

(Total 12 marks)

**Q8.**

An elastic string has natural length  $l$  and modulus of elasticity  $\lambda$ . One end of the string is fixed and a particle of mass  $m$  is attached to the other end of the string.

- (a) When the particle hangs vertically in equilibrium, the extension of the string is  $\frac{l}{4}$ . Show that  $\lambda = 4mg$ .

(2)

- (b) The particle is then set into vertical motion and  $x$  is the displacement of the particle from the equilibrium position at time  $t$ . Assume that the string remains taut during the vertical motion.

- (i) Show that  $\frac{d^2 x}{dt^2} = -\frac{4gx}{l}$ .

(5)

- (ii) Hence find the period of the motion.

(2)

(Total 9 marks)

**Q9.**

A particle moves with simple harmonic motion on a straight line between two points  $A$  and  $B$ , which are 0.4 metres apart. The maximum speed of the particle is  $10 \text{ m s}^{-1}$ .

- (a) Show that the period of the motion is  $\frac{\pi}{25}$  seconds.

(4)

- (b) Find the speed of the particle when it is 0.04 metres from  $A$ .

(3)

- (c) The distance,  $s$ , of the particle from  $A$  at time  $t$  is given by

$$s = p - q \cos(\omega t)$$

where  $\omega$ ,  $p$  and  $q$  are constants.

- (i) State the values of  $\omega$  and  $q$ .

(2)

- (ii) When  $t = 0$  the particle is at  $A$ . Find the value of  $p$ .

(2)

(Total 11 marks)

### Q10.

A particle attached to a spring moves with simple harmonic motion. The particle moves between the points  $A$  and  $B$ , which are 0.1 m apart. When the particle is 0.01 m from  $A$  its speed is  $0.6 \text{ m s}^{-1}$ .

- (a) Show that the period of the motion is  $\frac{\pi}{10}$  seconds.

(6)

- (b) Find the speed of the particle when it is at the midpoint of  $AB$ .

(2)

- (c) Find the magnitude of the maximum acceleration of the particle.

(2)

(Total 10 marks)

### Q11.

A particle, of mass  $m$ , is attached to one end of a spring. The spring has natural length  $l$  and modulus of elasticity  $\lambda$ . The other end of the spring is fixed to the point  $Q$ . Initially the particle is in equilibrium, vertically below  $Q$ .

- (a) Find, in terms of  $m$ ,  $g$ ,  $\lambda$  and  $l$ , the extension of the spring when the particle is in equilibrium.

(2)

- (b) The particle is then pulled down, so that it is vertically below its equilibrium position, and released. The displacement of the particle from its equilibrium position is  $x$  at time  $t$ . Show that

$$\frac{d^2 x}{dt^2} = -\frac{\lambda x}{ml}$$

(5)

- (c) Find, in terms of  $\lambda$ ,  $l$  and  $m$ , the period of the subsequent motion.

(2)

- (d) The spring is now replaced with another spring of natural length  $l$ , but with modulus

of elasticity  $\frac{\lambda}{4}$ . What happens to the period of the motion?

(2)

(Total 11 marks)

**Q12.**

A particle moves on a straight line. At time  $t$  the displacement of the particle from a point  $O$  is  $x$ . The particle moves with simple harmonic motion, so that

$$\frac{d^2x}{dt^2} = -25x$$

The particle is initially at rest at the point  $P$ , which is a distance  $a$  from  $O$ .

- (a) Find the time that it takes the particle to move directly from  $P$  to  $O$ .

(3)

- (b) Find, in terms of  $a$ , the distance of the particle from  $O$  when its speed is half its maximum speed.

(5)

(Total 8 marks)